

Is There an Upper Limit to Computer Processing Speed?

An Evidence-Bounded Research Proposal for Physical Compute Envelopes

Anonymous Research Proposal
Survey to Idea to ExperimentDesign to Author Workflow
IEEE Conference Format Research Plan

Abstract—There are physical limits to computer processing speed, but no single technology-independent number answers the question for every computer and every workload. A processor clock, transistor count, physical gate rate, sampling rate, and application time-to-solution have different denominators. They cannot be substituted for each other without an explicit computation boundary. This paper proposes the Physical Compute-Envelope Contract, a research protocol for evaluating a useful-throughput claim under a declared workload, correctness target, reliability target, architecture, energy, thermal environment, spatial extent, communication topology, memory assumptions, and input-output boundary. The proposal combines quantum speed limits, finite signal propagation, thermodynamic erasure cost, heat removal, information storage, and fault-tolerant logical overhead as separate constraint cards. It rejects two common overclaims: that the end of Moore scaling supplies a universal physical ceiling, and that quantum computation removes physical speed limits. Quantum processors can change algorithmic complexity for specified tasks, but state preparation, control, communication, measurement, error correction, and verification remain part of useful end-to-end computation. The resulting design is a falsifiable and architecture-neutral way to decide whether a proposed speed statement is supported within scope, incomplete, overgeneralized, or inconclusive. This is a design-only study: it reports no executed hardware experiment, simulation, benchmark, or observed performance result.

Index Terms—physical limits of computation, quantum speed limit, Moore’s Law, Landauer principle, fault-tolerant quantum computing, computer architecture, throughput, reproducibility

I. INTRODUCTION

The statement that computers will eventually become as fast as physics allows is compelling but underspecified. What counts as “fast”? A contemporary processor may advertise a clock frequency, an accelerator may advertise operations per second, a high-performance system may advertise floating-point throughput, and a quantum processor may report physical-gate performance, sampling throughput, or a task-specific computational advantage. None of these numbers independently specifies the useful work delivered to an end user. The computation could differ in problem size, output representation, accuracy, probability of success, initialization, readout, energy boundary, or communication path.

The historical motivation is clear. Moore’s influential observation concerned the number of components that could be integrated economically over time [1]. Earlier idealized scaling also tied smaller field-effect devices to favorable density, speed, and power trends [2]. Those relationships enabled extraordinary engineering progress, but they are not laws that a physical computer must obey. Leakage, variability, interconnects, power delivery, cooling, packaging, manufacturing cost, and architecture change the realized trajectory. The reported slowing of Moore-style scaling is therefore a reason to ask a physics question, not an answer to it [3].

Quantum information processing sharpens rather than removes the question. Quantum speed limits constrain how quickly specified quantum states can evolve under stated physical resources. Quantum algorithms can outperform classical alternatives for particular problem structures. Yet neither result creates a general-purpose, unbounded operations-per-second machine. A useful computation remains embodied in a device with finite energy, finite size, finite signal velocity, control hardware, thermal interfaces, memory, measurement, and a required probability of correct output.

This report simulates the repository workflow

Survey $\longrightarrow$ Idea $\longrightarrow$ ExperimentDesign $\longrightarrow$ Author. (1)

It gives a direct answer: *there are upper limits, but they are conditional physical compute envelopes rather than one universal speed constant*. The proposed contribution is not a new numerical limit. It is a testable method for stating which limit is active for one clearly defined claim.

II. SURVEY: WHY A SINGLE SPEED NUMBER FAILS

A. *A metric has a numerator, denominator, and scope*

Table I separates quantities often called computer speed. They should be compared only after their numerator, time boundary, and acceptance rule have been aligned. A clock is a periodic control signal; it is not an accepted answer rate. A physical quantum gate rate is not a logical algorithm rate. A sampling rate is not necessarily a rate for a practically useful task. Conversely, a slow physical operation may contribute to

TABLE I
COMMON SPEED METRICS AND THEIR MISSING CONTEXT

Metric		What it records	What it does not establish
Clock frequency		Local timing cadence	Useful work, memory behavior, energy, or I/O cost
Transistor count		Integration density	Architecture, yield, heat, or algorithmic work
Physical gate rate		Device-level operation time	Logical reliability and full algorithm completion
FLOPS or TOPS	or	Chosen arithmetic operation rate	Accuracy, data movement, and workload relevance
Sampling rate		Samples from a named distribution	General-purpose advantage or application utility
Time-to-solution		End result for one workload	Transfer to a different workload or condition

a faster algorithm if it changes computational complexity for the named workload.

The scope problem can be made explicit. Let W denote a workload including its inputs and output representation, ϵ a correctness or distributional-error target, and ρ a reliability target. A claim about useful throughput is meaningful only after it states an end-to-end time boundary. The direct dependent quantity in this proposal is

$$R_{\text{accepted}} = \frac{N_{\text{accepted}}}{T_{\text{wall}}}, \quad (2)$$

where N_{accepted} is the number of outputs meeting the declared ϵ and ρ criteria and T_{wall} includes or explicitly excludes preparation, communication, readout, verification, and classical postprocessing. Equation (2) is a definition, not a claim of measured performance.

B. Moore scaling is an engineering trajectory, not a terminal theorem

Moore’s original article was a projection about component density and cost-effective integration [1]. Dennard and colleagues analyzed a device-scaling regime in which electric fields and device dimensions were coordinated [2]. The importance of these sources is conceptual: they show why density, power, and speed were historically coupled. Their importance is not that either predicts an unavoidable absolute speed for future computers.

At small feature sizes, multiple bottlenecks decouple. A switching device may be fast while a wire, cache miss, package, analog interface, or thermal constraint determines wall-clock completion. Three-dimensional integration, special-purpose acceleration, better algorithms, reversible logic, optical interconnects, and heterogeneous systems can change the active bottleneck. The appropriate inference from an atomic-scale feature constraint is that specific integration routes have limits; it is not that every information processing architecture has reached one scalar ceiling.

C. Physical bounds constrain different mechanisms

Quantum speed limits offer a strong example of a real physical constraint that is often overextended. For an ideal closed quantum system evolving to an orthogonal state, the Mandelstam-Tamm and Margolus-Levitin forms can be written as

$$\tau_{\text{QSL}} \geq \max \left\{ \frac{\pi \hbar}{2\Delta E}, \frac{\pi \hbar}{2(E - E_0)} \right\}, \quad (3)$$

where ΔE is the energy standard deviation, $E - E_0$ is mean energy above the ground-state reference, and $\hbar$ is reduced Planck’s constant. The review by Deffner and Campbell organizes these bounds as limits on quantum evolution and control [5]; the original Margolus-Levitin result provides a related dynamical speed statement [4]. Lloyd’s physical-computation analysis provides a wider resource-bound perspective without making energy the sole whole-system denominator [7]. Equation (3) says neither that every quantum gate saturates the bound nor that the reciprocal of this time is a complete computer rate.

Finite transmission produces a different lower bound. If the critical information path has length L and signal velocity v_{sig} , then

$$\tau_{\text{signal}} \geq \frac{L}{v_{\text{sig}}}, \quad v_{\text{sig}} \leq c, \quad (4)$$

with c the speed of light in vacuum. The relevant velocity can be lower in a medium, and a real path also contains serialization, switching, and controller delay. This term applies to a named topology; it is not a replacement for a device-transition model.

Landauer’s principle addresses logically irreversible erasure. At temperature T , erasing n_{erase} independent bits has an ideal lower heat cost

$$Q_{\text{erase}} \geq n_{\text{erase}} k_B T \ln 2, \quad (5)$$

where k_B is Boltzmann’s constant [6]. The assumptions matter: (5) is not a charge for every logical operation and not a measured package power. It is nevertheless indispensable when a claim relies on repeated reset, irreversible control, measurement, or data disposal. Finite-time operations and heat extraction can add substantial cost above this ideal floor.

D. Quantum advantage requires task semantics

Quantum information can deliver algorithmic advantages, but the predicate is not “uses qubits.” Preskill describes the central role of noise and circuit depth for near-term quantum systems [9], while DiVincenzo emphasizes the physical requirements for initialization, control, measurement, and communication [8]. Random-sampling studies clarify that a computational-advantage claim depends on a task distribution, classical comparator, verification method, and complexity assumptions [10].

Photonic boson-sampling demonstrations are scientifically important task-specific evidence. They should not be reported

as a universal rate at which arbitrary programs become faster. A particular classical-runtime estimate can change with improved algorithms and hardware; the sampling task may have no direct application utility; and a universal, fault-tolerant machine has additional preparation, error-handling, and readout requirements. The survey therefore retains sampling as an architecture and workload card, not as proof that an upper speed limit vanished.

III. IDEA: THE PHYSICAL COMPUTE-ENVELOPE CONTRACT

A. From a broad question to a falsifiable object

The Idea Agent selected the PEC after rejecting a clock-only comparison, an energy-only ceiling, and a quantum escape narrative. The research object is one tuple

$$\mathcal{C} = \langle W, \epsilon, \rho, A, E, T, L, \mathcal{N}, \mathcal{H}, \mathcal{M}, \mathcal{I} \rangle, \quad (6)$$

where A identifies hardware and algorithmic architecture; $\mathcal{N}$ identifies communication topology; $\mathcal{H}$ identifies the cooling and heat-extraction boundary; $\mathcal{M}$ identifies memory and information-density assumptions; and $\mathcal{I}$ identifies input-output, observation, and verification boundaries. The other symbols are defined in Section II. A missing field is not a nuisance parameter; it is an unresolved part of the speed claim.

The PEC central hypothesis is that a layered contract distinguishes supported from unsupported useful-throughput statements more reliably than one component metric. It can be falsified. The framework fails if its complete card set cannot discriminate a well-supported claim from an unsupported claim, if a mandatory card adds no decision value across the declared workload family, or if independently verified logical throughput violates the stated envelope without a changed contract or accounting correction.

B. Compositional envelope rather than a false universal theorem

For a given contract, PEC defines the following reporting envelope:

$$R_{\text{accepted}} \leq \min\{R_{\text{transition}}, R_{\text{signal}}, R_{\text{thermal}}, R_{\text{memory}}, R_{\text{reliable}}, R_{\text{I/O}}\}. \quad (7)$$

Equation (7) is deliberately operational. It does not assert that the six right-hand terms are independent, tight, or exhaustive physical theorems. Instead, it prevents the more common error of reporting a favorable numerator while silently omitting a limiting system stage. A card may be marked inapplicable with a documented rationale; it may not be silently assumed favorable.

Table II specifies the minimum information in each card. For example, a reported quantum physical-gate time belongs in the transition card. It cannot establish logical useful throughput until the reliability card states encoding, error model, syndrome cycles, decoder, retries, and accepted-output criterion. A high-speed chip-to-chip link belongs in the signal card. It

cannot erase I/O cost if data preparation or output verification crosses a slower boundary.

IV. EXPERIMENTDESIGN: A DESIGN-ONLY EVALUATION PROTOCOL

A. Research unit and variable map

The ExperimentDesign Agent routes PEC to a computational-digital and mathematics-theory template. The unit of analysis is

$$U = \langle W, \epsilon, \rho, A, \tau_{\text{transition}}, \tau_{\text{signal}}, \mathcal{H}, \mathcal{M}, \Omega_{\text{logical}}, \mathcal{I} \rangle, \quad (8)$$

where Ω_{logical} is a physical-to-logical space-time and control-overhead description. The independent variables are architecture class and workload class. The dependent variable is R_{accepted} from (2). Controlled conditions include the correctness definition, reliability target, output representation, and time boundary.

The protocol cannot report a scalar readiness score. A scalar would hide whether an attractive transition time was paid for by an implausible cooling assumption, an omitted decoder, a changed output distribution, or an unreported classical preprocessing step. Instead it provides traceable cards and conditional decisions.

B. Pre-registered reasoning obligations

First, every transition claim must say whether (3) applies. A QSL may be inapplicable to an open, driven, nonorthogonal, or insufficiently specified process. Inapplicability is a valid result of the card, but substituting a reciprocal gate time for the QSL is not.

Second, the transport card must list the longest critical information path and the relevant medium. Equation (4) is only a propagation floor; queueing, routing, synchronization, and decoder feedback can be additional terms. A small local quantum processor can therefore be control-limited, while a large conventional processor can be memory- or network-limited.

Third, the thermal card separates a fundamental lower bound from a realized engineering budget. Equation (5) applies to declared irreversible erasures. A future study would document which resets, measurements, and control records count, then compare their dissipation and heat removal with the package boundary. Reversible logic is a crucial counterexample: it can change the erasure accounting, but it does not make signal latency, finite device volume, or error handling disappear.

Fourth, the reliability card converts physical process counts into accepted logical work. A useful reporting relation is

$$R_{\text{logical}} = \frac{R_{\text{physical}} Y_{\text{accept}}}{\Omega_{\text{space}} \Omega_{\text{time}} \Omega_{\text{control}}}, \quad (9)$$

where Y_{accept} is the probability that a result passes the declared acceptance rule, and the three Ω terms are documented resource multipliers. Equation (9) is a bookkeeping model, not a claim that all architectures factor exactly this way. It exposes why raw physical-gate speed is insufficient for a reliable algorithm.

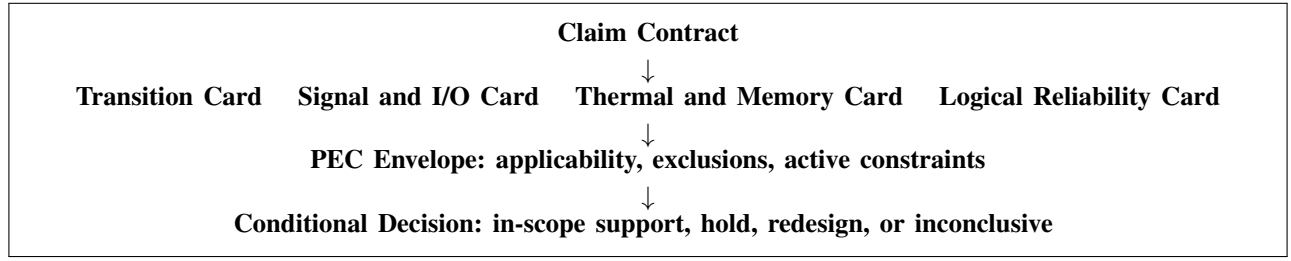


Fig. 1. PEC claim-to-decision architecture. The delivered editable Draw.io source expresses the same relationship. The diagram is a design artifact, not an executed benchmark.

TABLE II
PEC CARDS REQUIRED BEFORE A USEFUL-THROUGHPUT CLAIM

Card	Minimum evidence record	Claim invalidated when absent
Workload semantics	Input distribution, output form, accuracy metric, reliability threshold, wall-clock start and stop.	Different computations are treated as the same unit of work.
Transition	Switching or state-evolution model, control assumptions, and QSL applicability rationale.	A device-local time is promoted to a system rate.
Signal and I/O	Critical paths, topology, media, controller paths, initialization, readout, and verification inclusion.	Data movement or observation is silently excluded.
Thermal and memory	Irreversible operations, temperature, heat-removal limit, density assumptions, and finite-time losses.	Ideal erasure cost is reported as realized dissipation.
Logical reliability	Physical error model, code, space-time overhead, decoder, postselection or retry rule, and output acceptance probability.	Physical operations are counted as logical answers.
Transfer and comparator	Classical comparator, resource normalization, architecture version, and conditions of transfer.	A task-specific result is framed as universal speed.

TABLE III
FUTURE EVIDENCE MODALITIES AND THEIR BOUNDARIES

Evidence modality	Can test	Cannot establish alone
Source-reviewed device record	Populated transition, topology, and environment fields	End-to-end accepted throughput
Fixed-workload benchmark	Scoped correctness and time boundary	Universal physical ceiling
QEC resource analysis	Logical overhead under one error model	Performance under another hardware model
Thermal/package assessment	Compatibility with a named heat boundary	Minimum cost of all computation
Independent replication	Transfer under the same contract	Transfer to an unlisted workload or scale

C. Future evidence and conditional branches

No experiment, simulation, device operation, or cloud job is performed in this report. Table III states what a future research program could test after qualified human review. It also prevents each evidence source from carrying a claim it cannot establish.

The protocol yields six statuses. **CONTRACT EVALUABLE** means that all cards and exclusions are explicit, not that a speed claim is true. **BOUND SUPPORTED IN SCOPE** means that future evidence supports the named envelope only under the declared contract. **RELIABILITY ACCOUNT**

INCOMPLETE holds a claim when physical operations have not been translated into accepted logical outputs. **THERMAL OR I/O BOUNDARY MISSING** requires a redesign when cooling, preparation, readout, or verification has been omitted. **QUANTUM ADVANTAGE OVERGENERALIZED** requires the task and comparator to be separated from a general processor claim. **INCONCLUSIVE** is an allowed outcome when evidence cannot discriminate the envelope.

V. WHAT QUANTUM COMPUTING DOES AND DOES NOT CHANGE

Quantum computation matters because it can alter the relationship between workload size and required computational resources. It does not matter because it removes the requirement for a resource boundary. The DiVincenzo criteria remain a useful implementation checklist: scalable qubits, initialization, long coherence relative to operations, a universal gate set, measurement, and communications are physical requirements, not abstract algorithm steps [8].

Error correction makes the distinction sharper. Surface-code research describes a practical route to large-scale fault tolerance while emphasizing the associated architecture [12]. More recent cross-validated work reports low-overhead fault-tolerant memory and alternative code families [13], [14]. These sources justify optimism about reducing overhead; they do not license the omission of overhead. The relevant performance object is a logical computation with a target failure probability, not a raw physical-qubit cycle.

Random sampling demonstrates a related lesson. The photonic experiment by Zhong and colleagues is an important milestone in quantum computational advantage for a defined sampling task [11]. The subsequent review literature explains why the task, verification procedure, and classical simulation assumptions are part of the meaning of such a claim [10]. PEC records those conditions. It does not discount the demonstration; it blocks an invalid transfer from a specialized sampling experiment to “quantum computers are unconstrained by processing speed.”

VI. EXPECTED OUTCOME BRANCHES AND SCIENTIFIC PAYOFF

The anticipated contribution is a decision discipline, not a forecast. Table IV pre-registers allowable interpretations. It makes clear that an evidence-complete card set can authorize specialist evaluation without declaring deployment readiness, and that a favorable bounded result does not predict a universal successor technology.

The scientific payoff is comparability. A conventional accelerator and a quantum processor need not have identical mechanisms to be compared fairly. They need compatible workload contracts. If their answers have different accuracy, success probability, I/O inclusion, energy boundary, or latency boundary, the correct result is not a leaderboard. It is a statement that the comparison is underdetermined.

PEC also creates useful negative knowledge. A claim may be technically impressive but nontransferable: a new gate may reduce $\tau_{\text{transition}}$ without changing τ_{signal} ; a better code may reduce Ω_{space} while increasing controller complexity; a sampling benchmark may establish a nonclassical distribution without establishing application time-to-solution. These are not failures. They identify the specific next research dependency.

VII. RISKS, LIMITATIONS, AND HUMAN REVIEW

The proposal has deliberate limits. First, a minimum-of-terms envelope can miss couplings between heat, error rate, control bandwidth, and architecture. The purpose of (7) is disclosure, not a proof that every real system is separable. A future implementation must model couplings or flag them as unknown.

Second, physical bounds depend on assumptions. The energy reference in a Margolus-Levitin argument, the state distance in a QSL, the temperature and logical irreversibility in a Landauer calculation, and the signal path in a latency analysis all require careful source review. A numerical value should never be inserted merely because a formula exists.

Third, algorithmic complexity and classical simulation methods evolve. Consequently, a claim of quantum advantage needs a stated classical comparator and date-specific method, not only a historical runtime estimate. A different classical algorithm can change the benchmark meaning without changing the quantum device.

Finally, qualified device physicists, computer architects, quantum-information researchers, thermal engineers, and reproducibility reviewers must assess any real parameterization.

They must verify source applicability, units, device version, error model, workload specification, software stack, measurement inclusion, and independent replication protocol. This report is not a safety certification, a hardware control procedure, or an empirical validation.

VIII. RESEARCH PRIORITIES ENABLED BY PEC

A. A three-family validation suite

The immediate research priority is not to search for a grand universal constant. It is to make upper-limit arguments comparable across three deliberately different workload families. The first family should be a conventional memory-intensive workload, for which data movement, cache hierarchy, interconnect, packaging, and heat extraction may dominate a local arithmetic rate. The second should be a highly structured numerical workload, for which a specialized accelerator may improve useful throughput even if its clock does not. The third should be a quantum workload with a formally stated input distribution, output verification criterion, and classical comparator. This suite prevents the protocol from being tuned to one hardware culture.

For each family, the baseline and candidate must share a stated output semantic. A conventional method returning an approximate scalar estimate cannot be compared directly with a quantum sampler returning a distribution unless both are mapped to an agreed accuracy and utility criterion. A future study must therefore publish the workload generator, instance class, error metric, reliability threshold, and end-to-end stop condition before comparing rates. PEC treats this publication requirement as a scientific control, not as administrative documentation.

B. Separate fundamental floors from engineering ceilings

The proposed program also separates three kinds of statement that are often mixed together. A fundamental floor is an ideal constraint derived under explicit physical assumptions, such as (3) or (5). An engineering ceiling is a system-specific limit set by a package, controller, interconnect, fabrication process, or cooling system. A benchmark outcome is a performance observation on one workload and one software stack. These layers can inform each other, but none should be reported as the other.

This distinction gives direct hypotheses to future experiments. If a device improvement lowers an observed time-to-solution, the resource cards can ask whether the improvement came from a faster transition, a shorter data path, a better compiler schedule, lower error-correction overhead, changed postselection, relaxed accuracy, or a changed time boundary. If the answer is not known, the result remains valuable but cannot be used to update a fundamental upper-limit claim. Conversely, if an engineering ceiling becomes active far above an ideal floor, research may target the named engineering constraint rather than portraying the gap as a failure of physics.

TABLE IV
PRE-REGISTERED CONDITIONAL INTERPRETATIONS

Status	Authorized future interpretation	Required action
CONTRACT EVALUABLE	The claim is sufficiently specified for a physical and architecture review.	Do not call it a measured speed or deployment approval.
BOUND SUPPORTED IN SCOPE	A named active constraint is consistent with future evidence for the named workload and architecture.	Preserve all exclusions and compare only compatible contracts.
RELIABILITY ACCOUNT INCOMPLETE	Raw operations cannot yet be interpreted as accepted logical work.	Add an error model, code, overhead, and acceptance criterion.
THERMAL OR I/O BOUNDARY MISSING	The system boundary omits a possible limiting process.	Add cooling, data movement, preparation, readout, or verification evidence.
QUANTUM ADVANTAGE OVERGENERALIZED	A specialized computational demonstration is being transferred beyond its task.	State the task, comparator, verification method, and missing utility bridge.
INCONCLUSIVE	The evidence does not select an active envelope term.	Report no direction for a numerical upper limit.

C. A claim-reporting contract for reproducible comparisons

Every future PEC report should contain a compact machine-readable contract in addition to prose. The required fields are the workload identifier, output type, input distribution, accuracy target, reliability target, processor and compiler version, physical topology, timing boundary, energy and thermal boundary, memory accounting, communication paths, logical-overhead model, classical comparator, and uncertainty interval. Each field should record one of four states: supported by a primary source, independently verified in the future study, inapplicable with rationale, or unknown.

The contract is intentionally strict about unknowns. A missing cooling specification does not become an implicit infinite heat sink. An omitted decoder does not become zero latency. A missing classical preprocessing step does not become free. This rule is particularly important for cross-platform claims, because the largest apparent speedup can result from moving a cost outside the denominator rather than from changing the computation itself.

D. Predictions that can be wrong

PEC produces several falsifiable predictions about research reporting. First, after I/O, reliability, and thermal cards are added, some raw device-rate rankings will change for useful end-to-end workloads. Second, the active constraint will often change as an architecture scales: a local transition term may dominate a small prototype while wiring, synchronization, error correction, or cooling dominates a larger system. Third, claims with the largest transfer distance from a specialized benchmark to a general-purpose conclusion will show the greatest number of unknown cards.

These predictions can be tested by independent analysts using public specifications, controlled workloads, and documented assumptions. A negative result would be informative. If PEC cards never change an interpretation or if their decision statuses disagree systematically with independently validated comparisons, the framework should be simplified or rejected. The proposal therefore makes its own methodology subject to the same evidence discipline it asks of computing-speed claims.

IX. CONCLUSION

Computer processing speed has physical upper limits, but the correct scientific answer is not one scalar measured in hertz, FLOPS, gates per second, or transistor count. A valid limit is conditional on what the computer is asked to compute, how correctness and reliability are defined, where wall-clock time begins and ends, and which physical resources are allowed. Moore-style scaling remains historically important but is not a law of physics. Quantum computing can alter algorithmic resource scaling for specified tasks but does not bypass state-evolution, signal, thermodynamic, thermal, information, reliability, control, and I/O constraints.

The Physical Compute-Envelope Contract converts that conclusion into a falsifiable research program. It requires a workload contract and six linked constraint cards before a useful-throughput claim can be evaluated. It thereby replaces two misleading narratives - a universal terminal silicon speed and an unconstrained quantum future - with a more direct research question: which constraint controls a clearly stated computation, and what evidence would show that the constraint has changed? This is a stronger basis for comparing architectures and for designing the experiments that must follow.

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